TC-Lab: an $n = 1$ teaching study

Tiers 0–2, researcher review draft

Prepared for Alex Dowling

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Tier	Goal	Status and result
0	Test whether one shared thermal model explains all twelve ENV2 records. Diagnose ambient treatment and residual covariance.	Complete. Input-related residuals remain; adequacy unresolved.
1	Fit one five-minute sine and one step. Quantify uncertainty and predict the other ten records.	Complete. Narrow parameter ranges; poor transfer.
2	Repeat with five further pairs, balancing three steps and three sine periods.	Complete. Calibration choice changes estimates and predictions.
3	Consider the remaining one-sine/one-step combinations after review.	Deferred for Alex's feedback.

Current interpretation. P1 gives narrow thermal parameter ranges but transfers poorly to the other ten records. Its zero-ambient predictions have pooled RMSE 2.22 °C, and 19.0% of recorded temperatures fall inside the nominal 95% Gaussian observation ranges. Allowing the ±2 °C ambient envelope raises that fraction to 58.5%. The pooled residual analysis also retains a command-related cycle in the thirty-second sine records. These findings limit the proposed universal mean/error description. The uncertainty calculations remain conditional on that working description and its resampling sensitivities.

1 Model, records and teaching sequence

We accept the same-device ENV2 provenance as the study assumption. There are nine sine records (three runs each at 30, 120 and 300 seconds) and three step records. Calibration aliases with duplicate data are excluded.

For experiment j, the measured command u_j is held constant from its recorded timestamp to the next update. We propagate the two-state model exactly:

$$\theta \dot{z}_j = u_j - z_j, \tag{1}$$

$$\tau \dot{T}_j = -T_j + T_{\mathrm{amb},j} + K z_j, \tag{2}$$

$$T_{\mathrm{amb},j} = T_{1,j}(0) + \delta_j, \qquad -2\,^\circ\mathrm{C} \le \delta_j \le 2\,^\circ\mathrm{C}. \tag{3}$$

The positive K, τ, θ are shared across calibration records. The initial conditions are $z_j(0) = 0$ and $T_j(0) = T_{1,j}(0)$; the first temperature is conditioned on and its residual omitted. Initial temperature is a reference, not an independent ambient measurement. The drift comparison replaces the constant correction by a linear function with both endpoints inside the same band. Endpoint bounds are an implementation choice.

The input-lag state is one stage of the delay chain; it produces a first-order lag rather than an exact dead time. Numerical fit bounds are $K \in [0.2, 1.5]\,^\circ\mathrm{C}/\%$, $\tau \in [5, 600]$ s and $\theta \in [0.01, 200]$ s. These are declared computational choices, not measured physical limits. A finite-box profile alone cannot prove global identifiability.

With a fixed command-to-power conversion c, a lumped physical interpretation is $K = \alpha c/(UA)$ and $\tau = C_p/(UA)$. An unknown shared constant α is absorbed into fitted K. Estimating α separately adds no temperature-fit flexibility when K is free. Independent calibration is needed to recover absolute UA or C_p; it is not needed to fit K.

The cold, zero-offset command transfer is

$$G(s) = \frac{K}{(1+\tau s)(1+\theta s)}.$$

Its two poles can be exchanged without changing that response. The assumed ambient channel acts through τ, so a nonzero offset can distinguish labels conditionally on that channel. We retain both optimization basins and report unlabeled slow/fast poles where useful; pole labels alone do not establish heater or sensor constants.

The teaching sequence is: fit the mean, inspect residuals, fit their temporal covariance, transform response and sensitivities, refit, then reassess uncertainty and predictions. A diagonal set of weights cannot remove temporal correlation.

2 Tier 0: pooled structure exploration

Goal. Determine whether one shared $n = 1$ description explains all twelve runs, and use discrepancies to assess ambient drift and error structure.

Ambient treatment	RMSE, °C	SSE reduction	At bounds
Constant offset	0.46574	—	7/12
Linear endpoints	0.43702	11.95%	9/12

These are ordinary least-squares fits. Their raw residual lag-one correlations remain 0.879–0.966 for constant offsets. The training reduction from drift does not settle the ambient question.

We compared IID, Ornstein–Uhlenbeck (OU), OU plus independent measurement noise, and Matérn-3/2 plus measurement noise under both ambient treatments. Each cell used 24 starts and an alternating generalized-least-squares (GLS) fit. Independent dense-covariance reconstruction checked all 96 selected run trajectories and likelihoods. Alternating GLS and joint likelihood agree within 1.2×10^{-6} in negative log likelihood.

Constant-offset error model	Raw RMSE, °C	Innovation lag-one range
IID	0.46574	0.879 to 0.966
OU	0.46709	−0.541 to −0.307
OU + measurement noise	0.48320	−0.239 to 0.238
Matérn-3/2 + measurement noise	0.47981	−0.186 to 0.241

The working choice is constant offsets and OU plus measurement noise. Its pooled estimates are $K = 0.71001\,^\circ\mathrm{C}/\%$, $\tau = 137.48$ s, $\theta = 18.37$ s and correlation length $\ell = 213.87$ s. The total marginal residual SD is $0.47877\,^\circ$C, split into $0.47012\,^\circ$C slow variation and $0.09062\,^\circ$C independent measurement variation. Correlated likelihood emphasizes a different part of the residual spectrum than OLS, so a larger raw RMSE can accompany a better likelihood.

This working choice is agent-selected for the requested six-pair analysis and awaits Alex's scientific review. OU alone leaves strongly negative lag-one innovations; adding measurement variance reduces that effect. The remaining cycle still needs a mean-model explanation or a better disturbance model.

With the same covariance family, linear drift lowers negative log likelihood by 6.05 while adding twelve parameters; nominal AICc is 12.00 higher than for constant offsets. This is a conditional comparison under an imperfect error model.

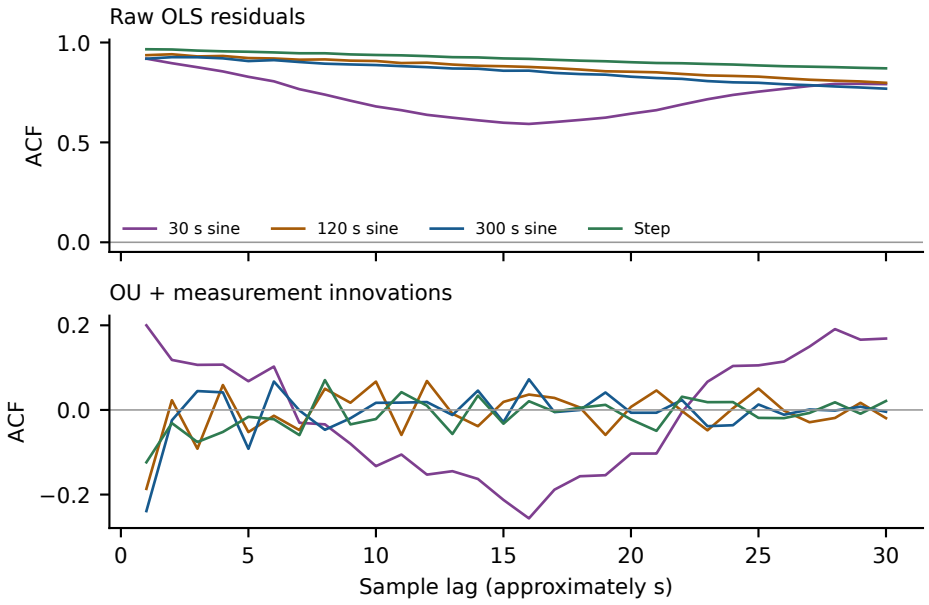


Figure 1: Residual correlation before and after the covariance-aware fit, using run 1 of each protocol. Whitening reduces the long persistence but leaves oscillation in the thirty-second sine and negative short-lag association in other records. These are descriptive ACFs; no IID significance bands are implied.

What remains in the residuals

The thirty-second sine runs have maximum absolute innovation/command correlations of 0.457–0.481 over the declared ±30-sample lag search. Widening ambient offsets to ±4 °C, changing startup by ±0.165 °C, adding one second of input prehistory, or using linear ambient drift leaves this cycle. Right-held commands worsen negative log likelihood by 256.30 and increase that association.

The temperatures lie on an inferred 0.32229984 °C grid after two-decimal reporting. A rounding sensitivity can reproduce some negative innovation correlation. It does not reproduce the command association. We generated and fully refitted 200 Gaussian and 200 rounded-measurement records from the pooled model. Neither mechanism reaches the observed maximum correlation of 0.481; their simulated 99th percentiles are 0.113 and 0.110. These plug-in diagnostic simulations challenge the working model; they do not identify the missing mechanism.

Ambient assumptions also affect interpretation. Widening offsets to ±4 °C removes active ambient bounds and lowers K by 2.47%, while the fast-sine discrepancy persists. Protocol-specific exploratory fits give $K = 0.655, 0.689, 0.724, 0.753$ for 30-second, 120-second, 300-second and step records. Those fits use different observations, so their likelihoods are not directly comparable and the point differences are not a formal heterogeneity test.

3 Tier 1: one sine and one step

Goal. Freeze the working structure, refit all numerical thermal, ambient and noise parameters using only five-minute sine run 1 and step run 1, then predict the other ten records.

The P1 point estimates are $K = 0.74063$ °C/%, $\tau = 147.60$ s and $\theta = 16.59$ s. Its two ambient corrections are 1.261 °C and 1.947 °C. No pooled noise estimate is fixed in this restricted fit.

Parameter	IID SD	HAC(30) SD	GLS SD	Full-curvature SD
K	0.0008118	0.003714	0.005981	0.00598
τ	0.1453	0.6578	1.235	1.232
θ	0.04781	0.2226	0.288	0.286

SD units are °C/% for K and seconds for the time constants. HAC changes the estimated OLS covariance while retaining its mean fit. IID and GLS use their own fitted means; their comparison includes the change in fitting criterion. GLS holds the covariance fixed. Full curvature includes noise estimation but remains a local, single-basin diagnostic.

Parameter	Gaussian 95% range	Block 95% range
K	[0.737, 0.757]	[0.737, 0.753]
τ	[145.900, 150.008]	[145.756, 150.088]
θ	[16.093, 17.266]	[15.947, 17.202]

Bootstrap refits include noise and ambient estimation, boundaries and pole alternatives. Percentile ranges summarize the working refit distributions; coverage has not been established. The opposite pole labeling occurs in 8/400 Gaussian and 3/400 block refits. A central interval can miss that secondary basin. The step ambient bound is active in 162/400 Gaussian refits. Nominal profile support is K: [0.7361, 0.7533]; τ: [145.7, 150.2]; θ: [16.02, 17.14]. Finite grid and crossing brackets are retained in the numerical archive.

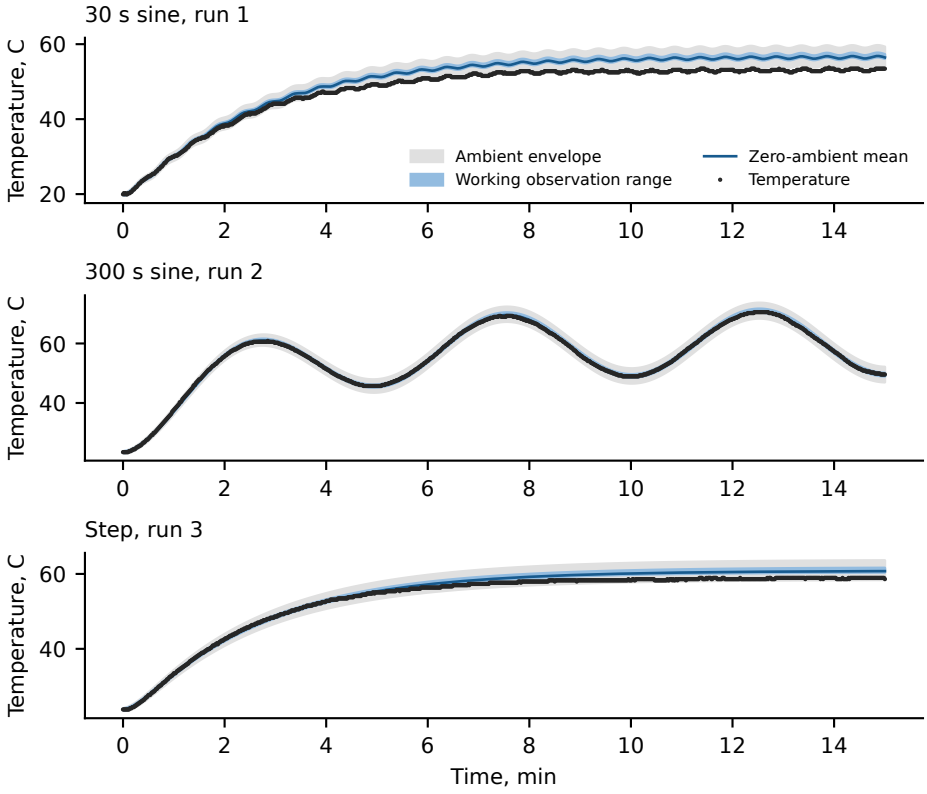


Figure 2: P1 predictions for three evaluation records. Blue bands are pointwise Gaussian working observation ranges at zero future ambient correction; gray envelopes also allow ambient offsets in ±2 °C. Ambient envelopes have no assigned probability. Black points are recorded temperatures.

4 Tier 2: five further pairs

Goal. Assess how much the result depends on the chosen sine and step. Six pairs in total use each step twice and each sine period twice.

Pair	Sine (period/run)	Step run	K, °C/%	τ, s	θ, s
P1	300 s / 1	1	0.74063	147.60	16.59
P2	30 s / 2	1	0.71402	127.70	22.17
P3	120 s / 1	2	0.72390	137.22	18.47
P4	300 s / 3	2	0.72193	146.61	16.12
P5	30 s / 3	3	0.67651	129.75	19.57
P6	120 s / 2	3	0.69734	130.62	18.87

All six point fits pass the declared stationarity checks and independent trajectory/likelihood reconstruction. Their parameter spread can reflect record variation, nuisance-bound effects and model inadequacy as well as statistical uncertainty.

Pair	Gaussian K range	Block K range	Swaps G/B
P1	[0.737, 0.757]	[0.737, 0.753]	8/3
P2	[0.691, 0.733]	[0.698, 0.728]	21/10
P3	[0.707, 0.736]	[0.711, 0.737]	1/1
P4	[0.718, 0.738]	[0.721, 0.739]	1/2
P5	[0.661, 0.705]	[0.661, 0.707]	0/0
P6	[0.686, 0.702]	[0.688, 0.702]	0/0

G/B counts use 400 draws per method. They are bootstrap selection frequencies under the working mechanism, not posterior probabilities of a physical pole assignment.

Pair	Unlabeled slow pole, s	Unlabeled fast pole, s
P1	[146.1, 150.7]	[16.1, 17.1]
P2	[117.5, 138.7]	[19.6, 24.6]
P3	[132.8, 140.6]	[17.9, 19.0]
P4	[144.9, 150.4]	[15.5, 16.7]
P5	[127.6, 131.8]	[18.3, 20.9]
P6	[129.3, 132.0]	[18.3, 19.3]

These are Gaussian percentile ranges. Unlabeled poles avoid combining two swapped parameter basins in a single interval for a physical time constant.

Pair	K width: 30-s blocks	60-s blocks	120-s blocks
P1	0.0147	0.0152	0.0163
P2	0.0293	0.0298	0.0481
P3	0.0257	0.0265	0.0215
P4	0.0215	0.0183	0.0180
P5	0.0447	0.0455	0.0460
P6	0.0148	0.0141	0.0127

Widths are in °C/%. Blocks count samples, approximately seconds. The 30/120-s sensitivities use 100 draws each; their 2.5% tails involve only two or three order statistics. The 400-draw primary methods have about ten draws per tail. Rare pole switches can move endpoints; the archive retains order-statistic Monte Carlo ranges and half-batch discrepancies.

Pair	Smallest information eigenvalue	Condition	Weak log direction
P1	2.77e+03	43.8	(-0.26, -0.36, 0.89)
P2	191	10.5	(0.13, -0.49, 0.86)
P3	2.04e+03	22.3	(0.54, 0.72, -0.44)
P4	1.5e+03	53.8	(-0.33, -0.42, 0.84)
P5	791	62.1	(-0.46, -0.11, 0.88)
P6	5.28e+03	15.5	(0.26, -0.24, 0.94)

Directions use $(\log K, \log \tau, \log \theta)$ after eliminating ambient columns; eigenvector signs are arbitrary. These are local fixed-covariance diagnostics and ignore box geometry and separated modes. All six conditional thermal information matrices have rank three at relative eigenvalue thresholds 10^{-8}, 10^{-10} and 10^{-12}. P2's full observed curvature is indefinite, so its regular Gaussian covariance is withheld. P4 has an active ambient bound despite positive observed curvature. Rank and positive curvature alone do not settle bounded or global inference.

Pair	OLS RMSE	GLS RMSE	Common sines	Ambient reconstruction
P1	2.175	2.219	2.168	0.930
P2	1.605	1.695	1.736	0.606
P3	2.038	1.950	1.831	0.629
P4	1.881	1.881	1.625	0.576
P5	1.853	2.223	1.431	1.085
P6	1.856	1.856	1.358	0.684

RMSE is in degrees Celsius. OLS/GLS columns pool each pair's ten evaluation records; the exclusions differ, so those aggregates are not a controlled same-test-set ranking. Common sines restrict GLS evaluation to 30-s run 1, 120-s run 3 and 300-s run 2, which are excluded from all six calibrations. Primary predictions use zero future ambient correction. The last column fits an offset using the complete evaluation response and is retrospective reconstruction.

Pair	Gaussian inside, %	Block inside, %	Gaussian + ambient, %
P1	19.0	20.5	58.5
P2	55.3	43.6	100.0
P3	32.9	29.3	89.8
P4	16.8	20.0	73.1
P5	50.8	54.2	80.7
P6	15.7	14.8	79.0

Fractions count measured temperatures inside pointwise future-observation ranges, pooled by sample count. Ambient envelopes are scenarios. Overlapping evaluation records and serial dependence prevent independent-binomial coverage claims.

P1's Gaussian K range is $[0.737, 0.757]\,^\circ\mathrm{C}/\%$; P5's is $[0.661, 0.705]\,^\circ\mathrm{C}/\%$. These conditional ranges do not overlap. The shared records, nuisance bounds and inadequate working error law prevent interpreting this as a formal independent test of physical gain variation.

P2 has the lowest ten-record aggregate prediction RMSE, 1.695 °C. P6 has the lowest RMSE on the common three sine records, 1.358 °C. Gaussian observation-range inclusion varies from 15.7% to 55.3% at zero future ambient correction, and from 58.5% to 100% with the bounded ambient envelope. Broader ranges can contain more observations while leaving the command-related discrepancy unresolved. GLS does not systematically reduce unweighted prediction RMSE across these pairs.

For teaching, P1 provides a direct example of narrow fitted parameter ranges accompanying poor transfer. The pooled analysis and the other five pairs then expose the mean/error assumptions behind those ranges. I recommend reviewing that shared description before extending the pair enumeration.

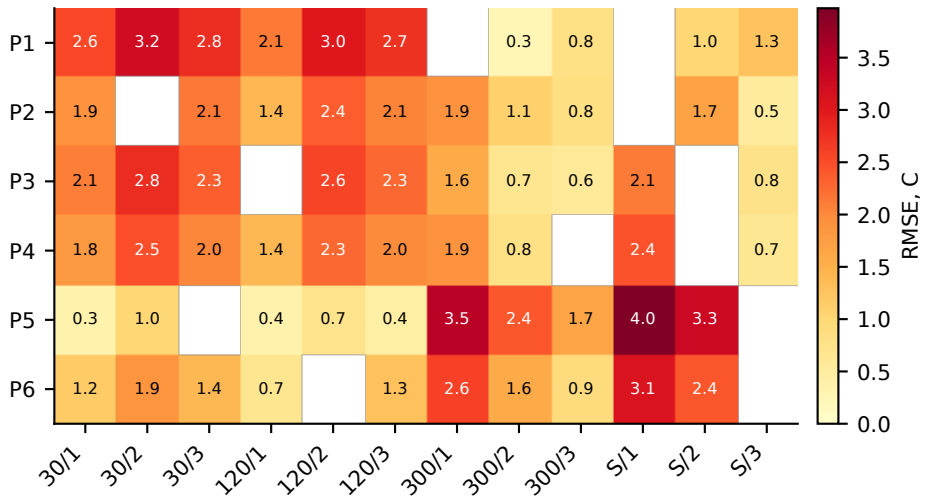


Figure 3: Zero-ambient prediction RMSE in degrees Celsius. White cells are the two calibration records for that pair. Records are labeled by sine period or step (S), then run number.

5 How uncertainty is calculated

For $e_j = y_j - m_j$, the working covariance is

$$C_j = \sigma^2 R_j, \qquad (R_j)_{ab} = (1-\eta)e^{-|t_a - t_b|/\ell} + \eta\, \mathbf{1}_{a=b}.$$

Here η is a variance fraction. A stationary innovations filter provides W_j with $W_j R_j W_j^{\mathsf{T}} = I$. We transform the response and every design column. For fixed nonlinear coordinates we solve bounded least squares for gain and ambient corrections, profile $\sigma^2 = \sum_j \|W_j e_j\|^2/N$, and minimize

$$-\log L = \tfrac{1}{2}\left[N\{\log(2\pi) + 1 + \log(\hat{\sigma}^2)\} + \sum_j \log|R_j|\right].$$

The determinant is required when estimating covariance parameters. State-space representations of half-integer Matérn kernels support exact temporal covariance computations with stationary initialization and irregular sampling (Hartikainen and Särkkä, 2010). They provide a computation method, not evidence that a chosen kernel describes these experiments.

The declared covariance search uses $\ell \in [0.05, 3000]$ s and $\eta \in [0, 1]$. Processes reset independently between experiments. Alternating GLS updates covariance from the current residuals, refits the transformed mean, and repeats; joint likelihood then checks the stabilized solution.

Local information uses $\log K, \tau, \theta$, removes ambient directions and reports eigenvalues/eigenvectors. We compare IID OLS, fixed-covariance GLS, Bartlett HAC sensitivities at 10/30/60 sample lags, and the observed likelihood curvature including noise estimation. HAC covariance addresses serial dependence under its assumptions (Newey and West, 1987). It cannot repair an incorrect mean. We withhold a regular Gaussian covariance when the observed curvature is indefinite; active bounds remain a qualification even when a matrix is positive.

We also profile each thermal parameter, refitting all remaining thermal, ambient and noise terms. Profiles can reveal asymmetric or extended likelihood support that a local quadratic approximation misses (Raue et al., 2009). The displayed $2\Delta(-\log L) = 3.84146$ cutoff is nominal; finite bounds, pole alternatives and error misspecification limit its coverage interpretation.

For every pair we completed 400 Gaussian parametric refits and 400 innovation-block refits with 60-sample blocks. Block-length sensitivities use 100 draws each at 30 and 120 samples. All nuisance and covariance parameters are re-estimated. Blocks resample centered standardized calibration innovations and are inverse-whitened. This custom sensitivity preserves short residual sequences but breaks alignment with the command; it cannot correct deterministic protocol bias. Finite simulation error, failed-refit denominators, pole-basin counts and bound hits are retained.

6 Prediction interpretation and open questions

Each pair drives ten other existing records using their commands and first temperatures. Primary predictions set the unknown future ambient correction to zero. Separate envelopes allow corrections anywhere in $\pm 2\,^{\circ}\mathrm{C}$; those envelopes do not assign a probability distribution to room temperature.

Mean-parameter ranges and future-observation ranges are separate. Observation ranges include estimated noise and parameter variation; they are pointwise, not simultaneous trajectory bands. A later ambient correction fitted to the complete evaluation temperature series is labeled retrospective reconstruction. It diagnoses level error but uses information unavailable to a prospective prediction.

Tier 0 informed structure selection using all twelve records. These evaluations are therefore retrospective checks of a selected model. The sixty pair/record evaluations overlap, and observations within each record are correlated. Empirical fractions inside a band cannot be interpreted as independent-binomial validation.

Before Tier 3, Alex's review should settle:

1. Whether the fast-sine command dependence warrants revising the mean before expanding the pair study.
2. Whether the $\pm 2\,^{\circ}\mathrm{C}$ ambient band is defensible, given the gain sensitivity and active bounds.
3. Which sine/step pair gives the clearest teaching sequence, including what its uncertainty fails to explain.
4. Whether to extend the same analysis to all 27 pairs or revise the working model first.

No new hardware experiments or Pyomo.DoE calculations are part of this phase.

Reproduction and evidence

Every job records its committed source, pinned scientific runtime and raw-data hashes. Independent matrix-exponential and dense-covariance calculations check propagation, likelihoods and whitening. The archive preserves solver failures, sensitivity choices and checkpoints. All 6,000 restricted-pair uncertainty refits were usable, with no failed or flagged draws. We regenerated 49 selected draws across every pair and mechanism, including pole switches when present. Their 24-start refits agree with the shorter refits within 6.9×10^{-10} negative log likelihood. This selected-draw check does not certify all optimization basins.

All eighteen thermal profile scans were independently checked at minima and crossing brackets. Five remote grid-point flags were reconciled using saved stationary attempts at the same likelihood to numerical precision; the original flags and dense derivative checks remain in the archive. An independent arithmetic audit checked all 336 bootstrap parameter distributions and all sixty prediction records.

Relevant source pages were inspected for all three method citations. The source/claim packet, physical-page records, citation inventory and export receipt accompany the report in the repository. These passages support the methods; the TC-Lab fit's scientific adequacy remains unresolved.

References

Jouni Hartikainen and Simo Särkkä. Kalman filtering and smoothing solutions to temporal gaussian process regression models. In *2010 IEEE International Workshop on Machine Learning for Signal Processing*, 2010. URL https://users.aalto.fi/~ssarkka/pub/gp-ts-kfrts.pdf. Author-hosted manuscript; physical PDF pages 2–3 and 5 inspected.

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